Attention Allocation and Navigation Information Design

For Newcomers to Hong Kong

Safari
SD5008 Research Design and Fieldwork Plan

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Research planning v2, 14 October 2026.

Research focus and question

How can navigation information be classified by its associated interaction burdens to inform interaction design methods for improving when and how it is presented and delivered?

Context

Unfamiliar walking journeys in Hong Kong using Google Maps.

Unit of analysis

One complete walk as a case, with information-interaction episodes inside it.

Expected output

A preliminary information classification and evidence-linked design guidance.

Knowledge gap and Literature review

OUR KNOWLEDGE GAP

In unfamiliar walking navigation, which information episodes and interactions add attention and interaction burden, which effectively support orientation, action and confirmation, and how do these differences vary with context?

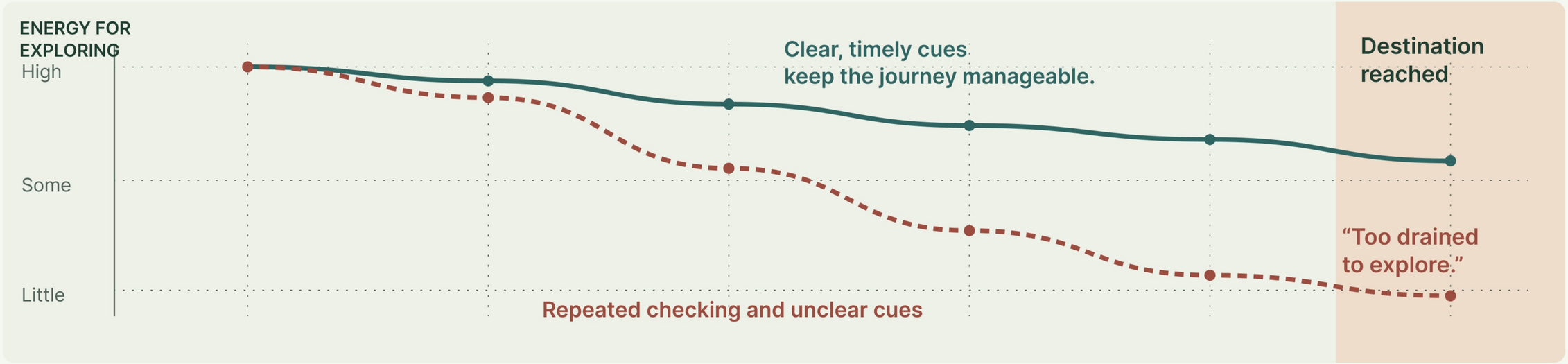
Literature review: Navigation supports arrival, but can still make it supportive or burdensome.

Study	What they tell us
Vaez et al., 2020	Digital navigation changes wayfinding strategies and does not necessarily reduce anxiety.
Wickens, 2008	Concurrent tasks can interfere when they draw on shared processing resources.
Oulasvirta et al., 2005	Urban mobile tasks fragment attention into brief interaction periods.
Cheng et al., 2023	Map information involves a trade-off between spatial learning and cognitive load in VR.
Golab et al., 2022	When pedestrians request instructions varies with environmental and personal factors.
Pielot et al., 2012	Tactile guidance can support eyes-free navigation in everyday use.
Shimokihara et al., 2025	Phone glances correlate with stops and errors; the order of events remains unclear.
Hardwicke et al., 2024	Wayfinding information can help users judge whether a walk is manageable.

Example:
A newcomer to Repulse Bay for holiday.

ILLUSTRATIVE USER JOURNEY

SET THE INTENT	NAVIGATE AN UNFAMILIAR ENVIRONMENT				START THE HOLIDAY
Plan a beach day Choose a beach.	**Find the right exit** Check map + signs.	**Follow street turns** Match street cues.	**Resolve unclear cues** Recheck directions.	**Confirm the beach entrance** Find beach access.	**Begin the holiday** Enjoy the beach.



Destination reached ≠ an effortless navigation experience.

Identify which information and interactions support the journey, and which consume attention and effort.

Paradigm, lens and method rationale

Pragmatism

The practical problem guides the evidence and method choices.
Context, consequences and participant values matter.

Theoretical lens

Attention allocation and task demands.
Multiple Resource Theory guides questions about possible competition.

Method

Focused qualitative field study.
Live observation addresses actions. Semi-structured interviews address explanations.

A phone glance or pause alone cannot establish cognitive overload. Six bounded cases cannot demonstrate a population effect.

Lee et al. (2026), W01, pp. 36–38, 42; Wickens (2008); Lazar et al. (2017), pp. 156–159, 174.

Why these methods?

Each method addresses a different part of the research question.

What we need to know	Methods / evidence	Why this fits
Actions and context	Live observation + timestamped fieldnotes	Capture the sequence of actions and surrounding conditions.
Goals and experience	Semi-structured interviews	Explore intentions and feelings that actions alone cannot explain.
Patterns and differences	Within-case analysis + cross-case comparison	Develop contextual information categories and design guidance.
Clips of specific moments for checking	Episode review using short clips, stills or notes	Connect participant explanations to particular navigation episodes.

Lazar et al. (2017), pp. 174–177, 188–189, 198–200; Russell & Chi (2014), pp. 373–379.

Research plan in detail

Next, we explain how we will carry out the study.

Participants and recruitment

6

planned main participants

Eligibility

- Adults who arrived in Hong Kong within six months
- Unfamiliar with the selected route
- Comfortable with the planned walk
- English or Mandarin interview

Recruitment and setup

- University and newcomer networks
- Record Google Maps experience, language and device
- Walk individually, without peer coaching
- Actual recruitment and completion: TBC

Participant target and eligibility are group choices. Lazar et al. (2017), pp. 175–176, supports a specified protocol.

Why start with six participants?

A time-bounded, exploratory study

Each case combines a walking observation with an in-depth interview.
Six is our initial target for a focused study, with limited generalisability.

01 Purposeful selection before the study

Select participants who meet our study criteria, so each case is highly relevant to unfamiliar walking navigation among newcomers to Hong Kong.

02 Feasible scope, with room for depth

Six cases fit the project schedule and allow detailed observation, interviewing, within-case analysis and cross-case comparison.

03 Iterative recruitment towards analytical sufficiency

If six cases provide insufficient evidence, recruitment will continue until interpretations are adequately supported or additional cases yield little substantive new insight.

Start Point: PolyU South Gate

Shared starting condition

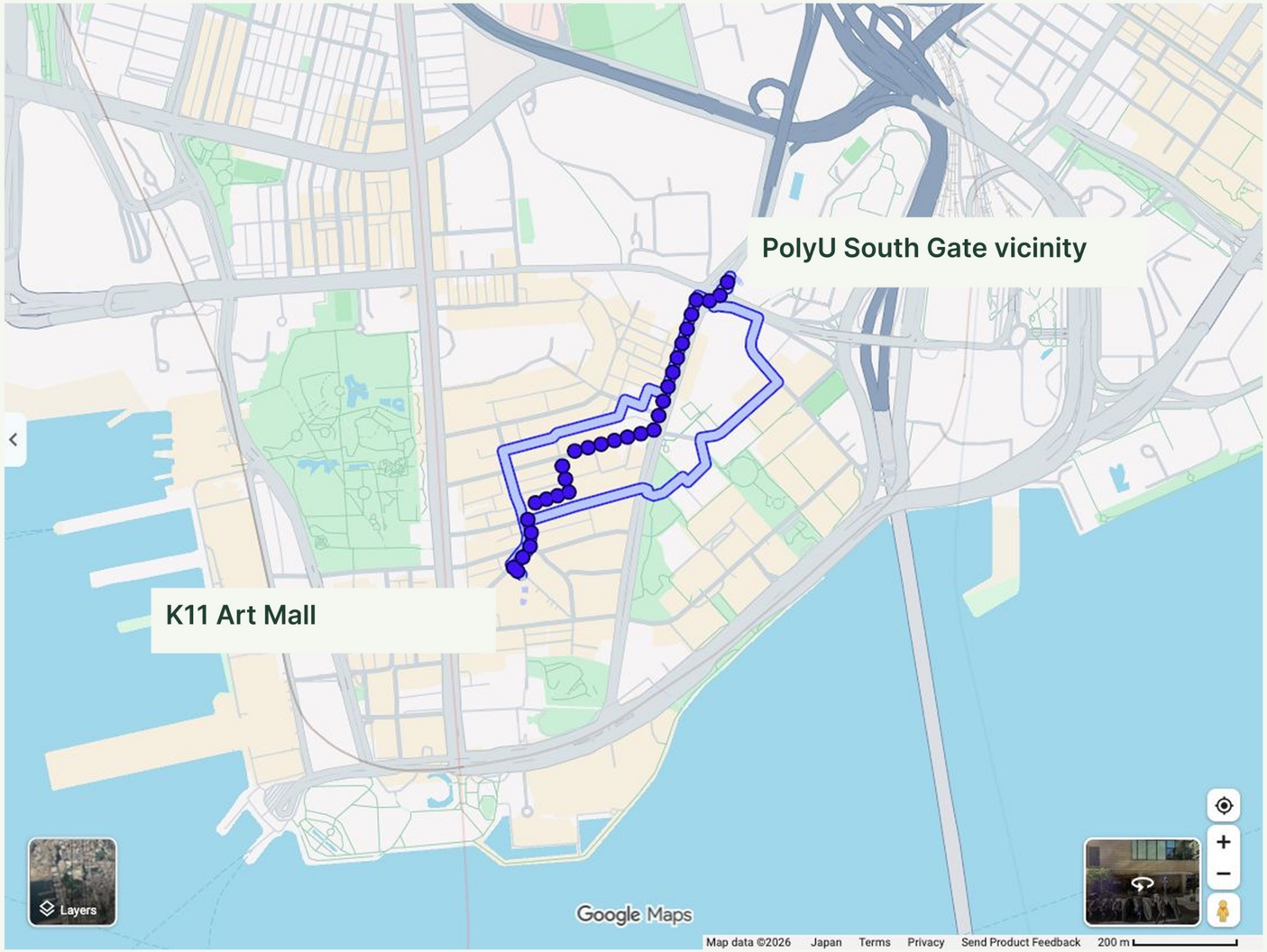
Main Entrance near Core A on Cheong Wan Road, matching the official campus map.

Meeting and T0

Agree a recognisable landmark outside the gate. Exact gathering point and on-site start marker: TBC.

PolyU Campus Development Office (n.d.); PolyU (2022). Campus map is a location reference, not a route scout.

Candidate A: K11 Art Mall



13 min / 900 m

Google Maps estimate

18 Hanoi Road, exterior entrance TBC

Mapped street sequence

Chatham Road South
Granville + Carnarvon Roads
Hanoi Road

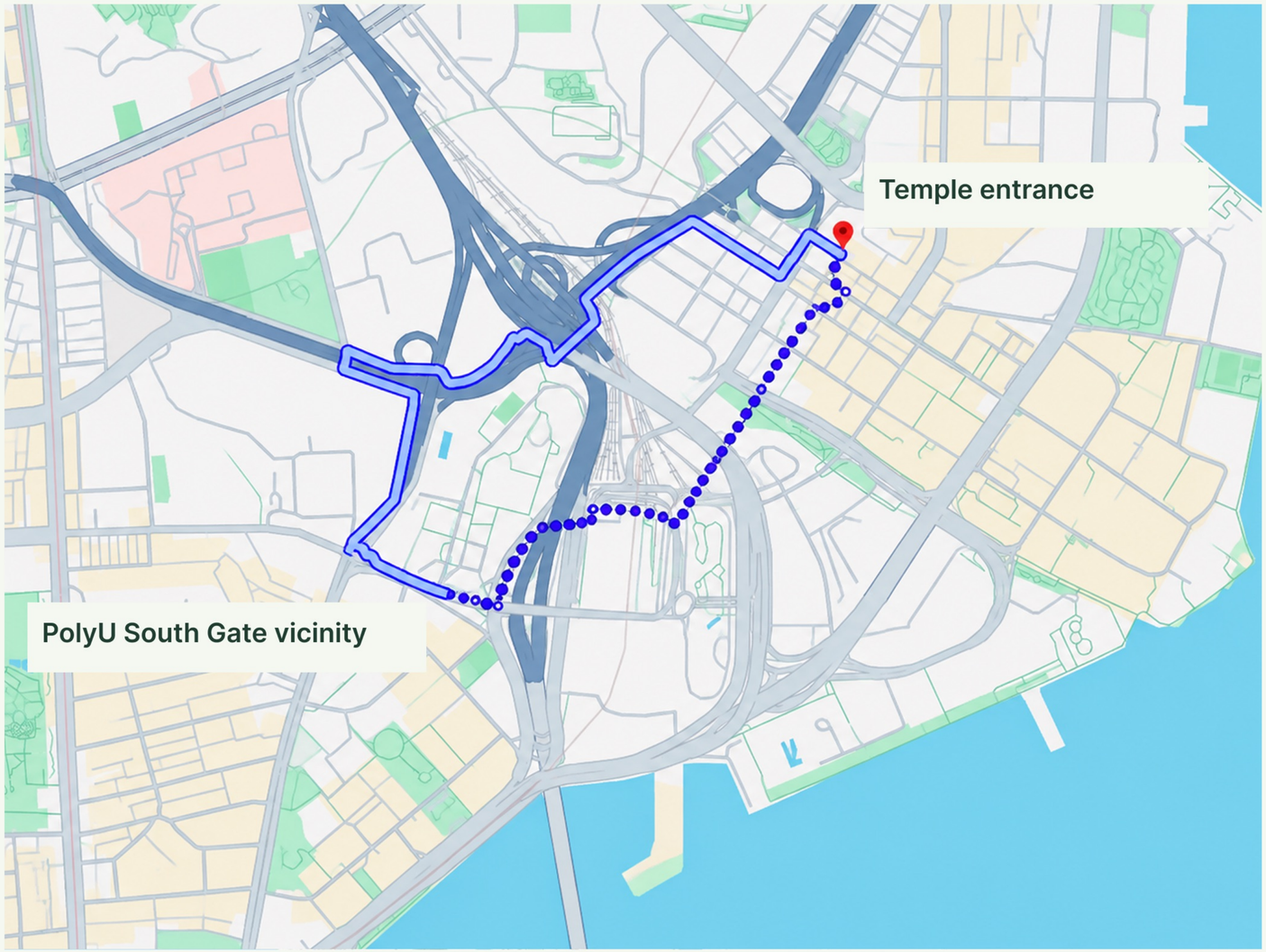
Observation opportunities to check

Street names, turn confirmation and destination-entry identification.

Screen this mall and approach explicitly. K11 Art Mall is different from K11 MUSEA.

Google Maps, 14 Oct 2026; K11 Art Mall (n.d.). Map estimate; scout TBC.

Candidate B: Hung Hom Kwun Yum Temple



Google Maps estimate; Chinese Temples Committee (n.d.). Scout TBC.

20 min / 1.3 km

Google Maps estimate

Station Lane, Hung Hom

Mapped street sequence

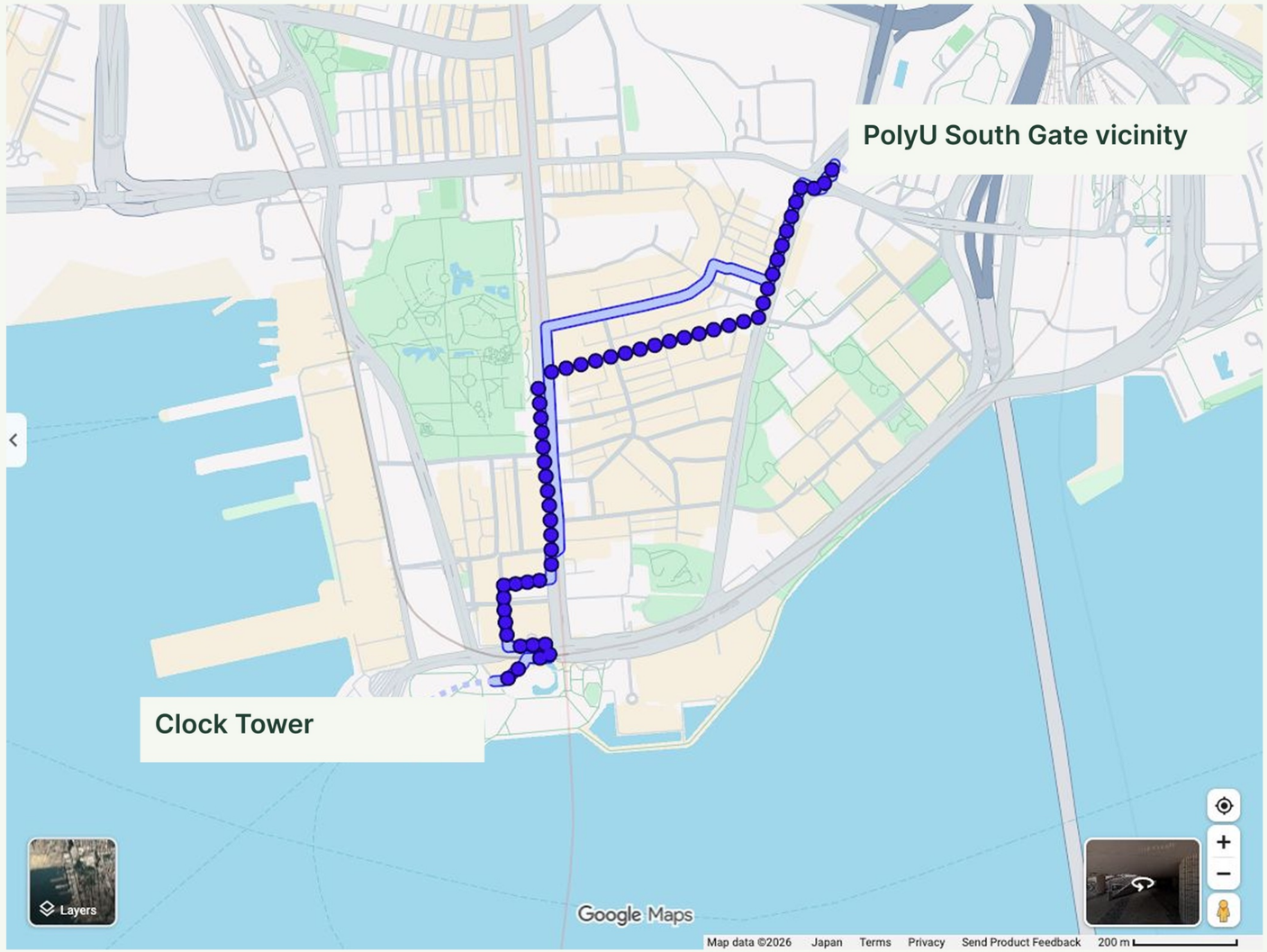
Chatham Road North
Hung Hom residential streets
Station Lane

Observation opportunities to check

Multiple turns, street orientation, and matching map directions with landmarks and the temple entrance.

Dense urban streets offer more navigation decisions. Check crossings, accessibility and temple entrance.

Candidate C: Clock Tower



24 min / 1.7 km

Google Maps estimate

10 Salisbury Road, exterior forecourt TBC

Mapped street sequence

Chatham + Granville Roads
Nathan + Hankow Roads
Salisbury Road to forecourt

Observation opportunities to check

Landmark finding and the street-to-waterfront approach.

Longer than the original 20-minute walk budget. Recheck crossings, resting space and session duration.

Google Maps, 14 Oct 2026; LCSD (n.d.). Map estimate; scout TBC.

Choose one common main route

All participants will follow the same fixed route during the main study.

Check route unfamiliarity

Ask about previous walks, exits and entrances before showing the route, no path teaching here.

If no route fits all participants

Recruit replacements or justify a route change within the approved protocol.

Compare the candidates below and confirm one fixed route in the pilot.

Candidate	Map estimate	Route considerations
A: K11 Art Mall	13 min / 900 m	Multiple turns through commercial streets. Prior familiarity may affect eligibility.
B: Hung Hom Kwun Yum Temple	20 min / 1.3 km	Dense urban streets with multiple turns and landmark recognition. Temple entrance requires checking.
C: Clock Tower	24 min / 1.7 km	Landmark finding and waterfront approach. Longer walking duration may require schedule adjustments.

Session flow and pilot budget

10 min

Briefing

Consent, background, phone setup and T0 procedure.

5–8 min

Warm-up

~300 m tutorial, procedure check and teach-back.

20 min

Main walk

Individual Google Maps task with live event notes.

25 min

Interview + debrief

Shared guide, selected episodes and wrap-up.

60–63 min + campus transfer (TBC). Pilot tutorial and total session timing. The Clock Tower map estimate exceeds the 20-minute main-walk allowance.

Group planning budget, not a prescribed book duration. Lazar et al. (2017), pp. 175–176, 210.

Researchers and participants warm-up

Warm-up for researchers

- Test route, recording, interview questions and total duration
- Two coders independently review the same excerpt
- Compare event boundaries, revise rules and version the codebook
- Keep outside the six main cases

Shared warm-up for each participant

- Same briefing + ~300 m unrelated practice
- Explain live notes, optional clips and observer roles
- Practise Google Maps without performance coaching
- Teach-back: task, safety, withdrawal and next steps

Keep practice outside main analysis. Prior app experience still differs; observer effects may remain.

Lazar et al. (2017), pp. 176, 210; Navalpakkam & Churchill (2014), pp. 339–340. Walking practice is our adaptation.

A shared understanding before the main walk

Standardise understanding before data collection

Establish a shared understanding of the task, observation and recording procedures to reduce procedural uncertainty and minimise avoidable influences on the data.

PROPOSED WARM-UP ROUTE (~300 m) / APP FIT: TBC

- **A** Library podium forecourt
- **B** University / Suen Chi Sun squares
- **C** Innovation Tower forecourt

1. Standardise the briefing

Explain task, roles, recording, safety and withdrawal.
Ask participants to explain the procedure back.

2. Practise the procedure

Walk with the observer and recorder; practise Google Maps.
Resolve procedural questions without navigation coaching.

3. Confirm the practice route

Check distance, access and app routing before use.
If Google routes off campus, revise or replace the practice.

4. Reset before the main task

Exclude practice records from the main analysis.
Log clarifications and restart timing at South Gate.

PolyU CDO campus map (detail); dashed trace = group proposal. Lazar et al. (2017), p. 176; Navalpakkam & Churchill (2014), pp. 339–340.

Field kit, roles and start checks

Field researchers

- Observer: actions + safety
- Recorder: events and time
- Log interventions: R_PROMPT
- No questions or device changes at crossings / stairs
- Interview: moderator + optional notetaker

Before T0

- Check consent and withdrawal
- Log device, map language and voice
- Sync HKT clock and elapsed timer
- Check battery, storage and recording scope
- Adjust only at a safe stopping point

Working materials

- Participant profile and familiarity screen
- Printed or digital event worksheet
- Current codebook and version
- Interview guide and episode list
- Camera: GoPro etc..;
- GPS apps: GTX Tracker etc...

Lazar et al. (2017), pp. 175–176, 210–212. Staffing and equipment are group planning choices.

Primary notes and auxiliary evidence

After each session, we will expand shorthand promptly. We will keep source disagreements visible and record observation coverage.

Primary	Timestamped human fieldnotes Observable action, context, uncertainty and researcher prompts.
Auxiliary	Consented short clips or stills Check selected details and cue interview recall. No exhaustive frame-by-frame coding.
Conditional	GPS context, only if approved and tested Position and timing support. Low speed does not identify STOP or attention.

Lazar et al. (2017), pp. 174, 211–212; Russell & Chi (2014), pp. 373–377. Notes-first emphasis reflects user-reported teacher advice.

Route and time recording

Navigate: Google Maps | Record: Open GPX Tracker (iPhone) / GPS Logger (Android) | Pilot: TBC

Interface example

01 Prepare

Check consent, device/OS and location access; wait for a GPS fix. Android: enable GPX export in Settings.

02 Record the formal walk

Start the GPX recorder → open Google Maps for navigation → mark T0. Keep the recorder running in the background, including stops.

03 Save and export

Mark the end first. iPhone: Pause → Save/name → Share GPX → Files.
Android: Stop → Tracklist → select track → Export / Share (GPX).

04 Verify and align

Use P01_MAIN.gpx; check coordinates, timestamps and gaps. Preserve raw UTC; align to HKT / T0 and the event sheet.

Pace is supplementary. Prioritise route, total time and observed events.

Merlos (n.d.), developer documentation; BasicAirData (n.d.), GPS Logger guide.

Observable action codebook

Seven initial action labels; co-coding is allowed. Causes and burden remain separate.

Code	Observable evidence	Boundary or exclusion
STOP	A visible halt in walking	Includes traffic, rest or photos. Context is required.
PHONE_VIEW	Discernible gaze at the phone	Holding a phone alone is insufficient.
PHONE_INPUT	Visible phone tap, swipe or other input	Do not infer which app or content without evidence.
HEAD_TURN	Visible head-direction change	Does not itself establish sign reading.
SIGN_VIEW	Evidence of viewing an identifiable sign	Use uncertain if the target cannot be established.
HELP_REQUEST	Audible or reliably noted request for assistance	Ordinary talk or ambiguous gestures are insufficient.
R_PROMPT	Researcher instruction or intervention	Log timing, content and reason.

New codes may be added as behaviours emerge in interviews, e.g. PHONE_SEARCH for searching for navigation-related information on a phone.

Group operational definitions. Coding principles: Wolff et al. (2024); Felner & Henderson (2022). Pilot calibration remains TBC.

CASE: STOP boundaries and context

1

Start

Mark the first visible halt.

2

During

Record context and overlapping codes.

3

End

Mark when walking resumes.

Hypothetical: 02:10 halt at red light. 02:12 phone gaze starts. 02:20 walking resumes.
STOP overlaps PHONE_VIEW. Traffic waiting is a plausible explanation.

Missed start/end: mark incomplete. Estimated time: mark approximate. Report gaps and coverage. GPS speed alone cannot define STOP.

Illustrative protocol example, not participant data or a quotation. Lazar et al. (2017), pp. 174–177.

Writable observation workbook

Blank preparation template. Complete actions live, then add the participant account and interpretation separately.

Event + time	Codes + raw observation	Context + source
Case: ____ Event: ____ Start (T0+): ____ End: ____ / approx.	Code(s): ____ What visibly happened: ____ Coverage / uncertainty: ____	Location / condition: ____ Fieldnote ID / clip time: ____ Coder / version: ____
Case: ____ Event: ____ Start (T0+): ____ End: ____ / approx.	Code(s): ____ What visibly happened: ____ Coverage / uncertainty: ____	Location / condition: ____ Fieldnote ID / clip time: ____ Coder / version: ____

Post-walk fields: participant account (quote / paraphrase / unknown) • interpretation • alternative explanation • follow-up needed. One row can carry multiple action codes.

Prepared from the group Word v2 codebook. Lazar et al. (2017), pp. 175–177, 211–212. This sheet contains no collected data.

Hypothetical evidence-chain example

Live observation

Example event E01:
STOP + PHONE_VIEW near a crossing.
Record timing, signal state and visible actions.

Participant account

Ask what they were trying to do.
Mark whether it is a verified quote, paraphrase or unknown.

Interpretation

Compare possible route confirmation with traffic waiting or another reason.
Support claim only with evidence.
Keep an alternative explanation.

No participant data appear here. A code identifies an action. The account and context support, challenge or leave its meaning unresolved.

Lazar et al. (2017), pp. 174–177; Russell & Chi (2014), pp. 373–377.

Camera recording to verify primary notes

Manufacturer product illustration
Equipment example only

Auxiliary verification only

- Check recorded event details and timing
- Primary notes remain the main record
- No separate video coding or analysis

Equipment example

An observer-worn action camera on a chest mount. GoPro Chesty is an equipment example; the final recording setup remains to be confirmed.

Before use

- Check consent scope, bystanders and site restrictions
- Pilot framing, comfort and start/stop controls
- Equipment, GPS and camera approval scope: TBC

GoPro (n.d.), official product illustration. Ethics receipt is user-reported. Approval conditions have not been inspected.

After walk interview: semi-structured

Whole walk	How would you describe the walk?
Information and sources	What information did you look for? Which sources did you use?
Next action	What helped you decide where to go next?
Helpful or difficult moments	Which moments were straightforward? Was anything unclear?
Optional design reflection	When or how would you prefer to receive that information?

Same core topics for all six. Order and follow-up can vary. Ask one question at a time, allow pauses and follow the participant’s own wording.

Lazar et al. (2017), pp. 198–200, 207–208. These prompts are group drafts, not book quotations.

Episode recap and interpretations

Select

3–5 logged episodes, including smooth or helpful moments.

Cue

Show a consented short clip or still. Use notes if no recording exists.

Ask

Do you remember this moment?
What was happening here?
What were you trying to do?

Accept “I don’t remember” and “it had no effect.”
Ask “Was anything unclear, or was this straightforward?”
Avoid assuming overload or difficulty.
Log cue source and any mismatch with live notes.

Russell & Chi (2014), pp. 373–379; Lazar et al. (2017), pp. 207–208. Recall can be reconstructive.

Analysis Framework: From Evidence to Design

1

Case record

Complete walk and event sources

2

Interpretation

Information needs, actions and participant accounts

3

Comparison

Patterns and differences within and across cases

4

Classification

Information roles, contexts and associated interaction burdens

5

Design guidance

Timing, content and interaction methods

Analysis dimensions: Information content, source, purpose, context, significance and associated interaction burden. Classification will consider both helpful interactions and difficulties, while design implications require further validation.

Lazar et al. (2017), pp. 176–177; Wickens (2008). Classification dimensions are provisional group choices.

Contingencies

Record changes before main sessions. Changes outside approved scope require tutor clearance before use.

Problem	Response	What to record
Too few eligible people	Widen recruitment within criteria. Replace familiar-route volunteers.	Eligibility, achieved N and justified changes.
Weather, works or unsafe access	Reschedule or use a previously checked backup.	Route change, weather, access and comparability.
Camera / GPS failure	Continue with consented live notes. Flag missing auxiliary coverage.	Missing segments, time offsets and source gaps.
Safety intervention / withdrawal	Pause or stop. Follow consent and approved handling.	R_PROMPT, reason and usable / excluded evidence.

Lazar et al. (2017), pp. 175–176. Teacher P2 guideline specifically requests participant-recruitment contingencies.

Actual progress and site readiness

Current preparation

- Fixed research question and pragmatism
- Six-person target and Google Maps condition
- South Gate start and three map candidates
- Codebook, workbook and interview guide

Execution status

- Selected route / exact T0 landmark: TBC
- Approval scope for camera and GPS: TBC
- Scout and independent pilot: TBC
- Recruited / completed main sessions: TBC

Actual documentation space: scout photo ID ____ • pilot worksheet ID ____ • dated protocol change ____ • participant / session count ____ . No fieldwork evidence has been inserted.

User-confirmed plan and Word v2. Do not treat preparation, map capture or an ethics receipt report as completed fieldwork.

Academic methods references

Methods used in the study; APA 7. Page locators refer to printed book pages.

Lazar, J., Feng, J. H., & Hochheiser, H. (2017). *Research methods in human-computer interaction* (2nd ed.). Morgan Kaufmann.

https://www.sciencedirect.com/book/9780128053904/research-methods-in-human-computer-interaction

Navalpakkam, V., & Churchill, E. F. (2014). Eye tracking: A brief introduction. In J. S. Olson & W. A. Kellogg (Eds.), *Ways of knowing in HCI* (pp. 323–348). Springer.

https://doi.org/10.1007/978-1-4939-0378-8_13

Russell, D. M., & Chi, E. H. (2014). Looking back: Retrospective study methods for HCI. In J. S. Olson & W. A. Kellogg (Eds.), *Ways of knowing in HCI* (pp. 373–393). Springer.

https://doi.org/10.1007/978-1-4939-0378-8_15

Case-study protocol, observation and interviews: Lazar et al. (2017). Retrospective recall: Russell & Chi (2014).

Navigation literature references (1/2)

Eight selected studies used on Slide 03 — alphabetical order, Cheng to Oulasvirta.

Cheng, B., Lin, E., Wunderlich, A., Gramann, K., & Fabrikant, S. I. (2023). Using spontaneous eye blink-related brain activity to investigate cognitive load during mobile map-assisted navigation. *Frontiers in Neuroscience, 17*, Article 1024583.
https://doi.org/10.3389/fnins.2023.1024583

Golab, A., Kattenbeck, M., Sarlas, G., & Giannopoulos, I. (2022). It's also about timing! When do pedestrians want to receive navigation instructions. *Spatial Cognition & Computation, 22*(1–2), 74–106.
https://doi.org/10.1080/13875868.2021.1942474

Hardwicke, J., Hill, K. M., & Ryan, D. J. (2024). Finding your way: Exploring urban park users' engagement with a wayfinding intervention through intercept go-along interviews. *Cities & Health, 8*(6), 1003–1016.
https://doi.org/10.1080/23748834.2024.2313307

Oulasvirta, A., Tamminen, S., Roto, V., & Kuorelahti, J. (2005). Interaction in 4-second bursts: The fragmented nature of attentional resources in mobile HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 919–928). Association for Computing Machinery.
https://doi.org/10.1145/1054972.1055101

Selected literature, not a systematic review. Findings are interpreted within each study's task and setting.

Navigation literature references (2/2)

Eight selected studies used on Slide 03 — alphabetical order, Pielot to Wickens.

Pielot, M., Poppinga, B., Heuten, W., & Boll, S. (2012). PocketNavigator: Studying tactile navigation systems in-situ. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 3131–3140). Association for Computing Machinery.

https://doi.org/10.1145/2207676.2208728

Shimokihara, S., Tabira, T., Ikeda, Y., Maruta, M., Han, G., Kamasaki, T., Hidaka, Y., Akasaki, Y., Kukizaki, W., & Kumura, Y. (2025). Association between smartphone glances during application-based navigation and pedestrian navigation performance: A real-world experimental study. *mHealth, 11*, Article 63.

https://doi.org/10.21037/mhealth-25-33

Vaez, S., Burke, M., & Yu, R. (2020). Visitors' wayfinding strategies and navigational aids in unfamiliar urban environment. *Tourism Geographies, 22*(4–5), 832–847.

https://doi.org/10.1080/14616688.2019.1696883

Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors, 50*(3), 449–455.

https://doi.org/10.1518/001872008X288394

Associations do not establish causation; our field methods do not directly quantify attention or cognitive load.

Coding and GPS technical references

Qualitative coding guidance and the documented tools used for route logging.

BasicAirData. (n.d.). *BasicAirData GPS Logger: Getting started guide*.

https://basicairdata.eu/projects/android/android-gps-logger/getting-started-guide-for-gps-logger/

Felner, J. K., & Henderson, V. (2022). Practical strategies for health equity researchers to enhance analytic rigor and generate meaningful insights from qualitative data. *Preventing Chronic Disease, 19*, Article 220134.

https://doi.org/10.5888/pcd19.220134

GPS.gov. (n.d.). *GPS accuracy*.

https://www.gps.gov/gps-accuracy

Merlos, J. M. (n.d.). *Open GPX Tracker for iOS + watchOS* [Mobile app and developer documentation]. GitHub.

https://github.com/merlos/iOS-Open-GPX-Tracker

Wolff, B., Mahoney, F., Lohiniva, A. L., & Corkum, M. (2024, August 8). *Collecting and analyzing qualitative data*. Centers for Disease Control and Prevention.

https://www.cdc.gov/field-epi-manual/php/chapters/qualitative-data.html

Coding labels are group definitions. Software instructions and background logging still require a pilot on each device.

Map and location sources

Location references and planning maps; route estimates shown are not field measurements.

Chinese Temples Committee. (n.d.). *Kwun Yum Temple, Hung Hom*. Retrieved October 15, 2026, from https://www.ctc.org.hk/en/temple/紅磡觀音廟/

Google. (n.d.). *Google Maps walking directions* [Maps]. Planning captures supplied for the three candidate routes, October 14, 2026. https://www.google.com/maps

K11 Art Mall. (n.d.). *Visit*. Retrieved October 15, 2026, from https://hk.k11.com/visit/

Leisure and Cultural Services Department. (n.d.). *Tsim Sha Tsui Clock Tower*. https://www.lcsd.gov.hk/en/hkcc/TSTClockTower.html

The Hong Kong Polytechnic University. (2022, August 24). *PolyU main entrance unveiling ceremony marks a key milestone in the University's history: Embarking on a new journey to education, research and knowledge transfer excellence*. PolyU official media release — 24 August 2022

The Hong Kong Polytechnic University, Campus Development Office. (n.d.). *Campus map* [Map]. https://www.polyu.edu.hk/cdo/docdrive/Useful%20Information/CampusMap.pdf

Google Maps captures retain their attribution. Route-query links appear in the notes for Slides 11–13.

AI and software record

Use in this revision

OpenAI Codex assisted with presentation layout and design, polishing words and sentences, and locating images.

Disclosure

Full AI-use documentation is retained with the project materials.

Next step

Confirm approval scope and the route on site, calibrate the recording rules in an independent pilot, then recruit and schedule the six individual main sessions.

AI-assisted preparation; group review remains required. No AI-generated fieldwork data.